

Semantic-Functional Communications in Cyber-Physical Systems

Pedro E. Gória Silva, Pedro H. J. Nardelli, Arthur S. de Sena, Harun Siljak, Niko Nevaranta, Nicola Marchetti, and Rausley A. A. de Souza

ABSTRACT

This paper explores the use of semantic knowledge inherent in the cyber-physical system (CPS) being studied in order to minimize the use of explicit communication, which refers to the use of physical radio resources to transmit potentially informative data. It is assumed that the acquired data have a function in the system, usually related to its state estimation, which may trigger control actions. We propose a semantic-functional approach that can leverage semantic-enabled implicit communication while ensuring the system maintains its required functionality and performance. We illustrate the potential of this proposal through simulations of a swarm of drones jointly performing remote sensing in a given area. Our numerical results demonstrate that the proposed method offers the best design option regarding the ability to accomplish a previously established task—remote sensing in the addressed case—while minimizing the use of explicit communication by controlling the trade-offs that jointly determine the performance and effectiveness of the CPS in terms of resource utilization. In this sense, we establish a fundamental relationship among energy, communication, and functionality, considering a specific end application.

INTRODUCTION

The fundamental limit of *explicit* communication, i.e., when a physical medium is used to transfer data from one point to another, was determined by Shannon [1]. Channel capacity was then defined as the maximum rate at which data can be transferred with an error probability approaching zero. Following the terminology of linguistics [2], channel capacity refers to the *syntax* of the communication problem (i.e., the explicit structure of “words” that constitutes what is a “well-formed sentence”). *Semantics* is different: it refers to meaning (i.e., the signification of “words”).

By shifting the domain from syntax to semantics, the communication problem changes from optimizing the use of the physical medium to when and how to use it most effectively, considering both the meaning and use of the data at the endpoint. In this case, the actual meaning of the data to be transmitted is an attribute of a given system, process, or

phenomenon. Following the conceptualization of [3], we can say that the semantics is immanent from the CPS in consideration, where data are acquired from a physical phenomenon to be then processed before being used in a decision-making process; all types of data processing refer to the cyber domain. Moreover, all data are acquired with certain goals in mind, like controlling an industrial process. This conceptualization is also supported in a broader perspective by [4], where it is stated that: *Shannon information theory provides various measures of so-called syntactic information, which reflects the amount of statistical correlation between systems. By contrast, the concept of ‘semantic information’ refers to those correlations which carry significance or ‘meaning’ for a given system.*

A novel approach called *semantic-functional communications* (SFC) was recently proposed in [5] and [6]. This method is more suitable for events in the context of machine-type communications like alarms, temperature monitoring, etc. Such a technique takes advantage of an event that has a meaning concerning the physical process and enables event-based transmissions. In most of the cases studied in these papers, SFC has outperformed other medium access control (MAC) protocols, such as Time-Division Multiple Access (TDMA) and slotted ALOHA, for massive sensor node scenarios.

The difference between the proposed SFC and other communication system designs is to design the physical layer in order to consider the “idle states” of given radio resources as informative to the receiver in semantic terms, in addition to providing an efficient method of accessing the medium based on the semantic understanding of the data. In other words, by not explicitly transmitting any data, the receiver “recovers” the information about the predetermined state of the system under consideration. This was tested in a multiuser scenario, where different sensors need to transmit only alarm messages to a decision-maker node. Such an initial case-specific solution motivated us to generalize this idea in order to characterize the fundamental limits of SFC.

Recent studies have addressed monitoring and surveillance using drones and other apparatuses. For instance, Kim et al. [7] proposed AI-assisted communications operating jointly

Pedro E. Gória Silva (corresponding author) and Niko Nevaranta are with the School of Energy Systems, Lappeenranta-Lahti University of Technology, 53851 Lappeenranta, Finland; Pedro H. J. Nardelli is with the School of Energy Systems, Lappeenranta-Lahti University of Technology, 53851 Lappeenranta, Finland, and also with the Centre for Wireless Communications, University of Oulu, 90014 Oulu, Finland; Arthur S. de Sena is with the Centre for Wireless Communications, University of Oulu, 90014 Oulu, Finland; Harun Siljak and Nicola Marchetti are with Trinity College Dublin, Dublin, D02 PN40 Ireland; Rausley A. A. de Souza is with the National Institute of Telecommunications (Inatel), Santa Rita do Sapucaí 37540-000, Brazil.

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with devices, including unmanned aerial vehicle (UAV), mobile robots, and smart devices, on the aerial and ground sides for surveillance and epidemic prevention. On the other hand, we exclusively address a challenge for future generations of wireless networks: controlling agents involved in monitoring as part of cyber-physical systems. Here, we propose SFC with event-based transmission for monitoring with drones, while recent literature employs fifth generation (5G), for instance [7], [8].

This article focuses on the fundamental communication limits of transmission of informative semantic data to perform well-determined functions within CPSs. The proposed approach consists of minimizing the use of explicit data transfer, i.e., maximizing “implicit communications,” by leveraging the “informative silences” of idle states.

BACKGROUND: SEMANTIC COMMUNICATIONS

Following the current trend, the technical literature does not provide a precise or standardized delimitation for the concept of semantic communications; it has dealt with this issue by distinguishing it from the syntactic (Shannon-like) approach to the communication problem. In simple terms, *syntactic* communications aim to establish a flow of information without taking into account any intrinsic quality of the data, while *semantic* communications explore the (preliminary) meaningful knowledge of the data that entities may previously have or are capable of acquiring.

However, a sharper distinction between these two theoretical classes requires more attention. We will discuss how the current literature on wireless communication deals with the “semantics” based on three main categories: *semantic-oriented communications*, *goal-oriented communications*, and *semantic-aware communications*. The reader is encouraged to peruse [9] to thoroughly understand the most recent developments in semantic communication theory and the reasons behind such a classification. Figure 1 presents a comparison between the different classes of semantic communication. In the following, we will provide a brief overview of the topic.

SEMANTIC-ORIENTED COMMUNICATIONS

A semantic-oriented communication system bases its coding process on the quantification of semantic information, i.e., by primarily using the *Theory of Strongly Semantic Information* [10], it aims to achieve semantic-level communications. One strand of this theory models the communication system through a fundamental type source that can make factual statements in propositional logic. The major distinction from Shannon’s information theory is that logical probabilities guide the semantic information measure. These logical probabilities are defined by background knowledge instead of statistical probabilities. Furthermore, one can consider the prior knowledge of the destination about the source in the coding

process. Therefore, the main difference between a semantic-oriented communication system and a traditional one is the data processing based on semantic characteristics.

GOAL-ORIENTED COMMUNICATIONS

In the last decades, research efforts have been made to study and clarify the concept of *goal-oriented communications* by defining *goals* (or *tasks*) in communication. On the one hand, we have goals that capture the intent of communicating parties, called *meta-goals*. On the other hand, we have goals that capture effects observed by a third party, named *syntactic goals*. Based on these definitions, some papers indicate that communications systems can accomplish meta-goals even if the two parties do not share the same language (i.e., protocol). To make this feasible, the third party must be able to assess whether the meta-goals have been achieved.

In recent works on goal-oriented communications, the aim is to take advantage of the growth of computational power in conjunction with state-of-the-art Artificial Intelligence (AI) techniques. While the communication principle—enabling the parties to perform collaborative communication aiming at a common goal or task—remains unchanged, the optimization of the task (the process performed by the parties in order to complete the task) is carried out by recently developed AI algorithms. One example is the task-oriented multiuser semantic communication system for multimodal data transmission proposed in [11]. There, the authors discuss how semantic transmission can be used to inquire information about images by correlating multimodal data from multiple users.

SEMANTIC-AWARE COMMUNICATIONS

Semantic-oriented and goal-oriented communications address a connection between two specific agents. Even though the network comprises multi-agents, the peer-to-peer interaction works as a basis for completing a task (meta-goal). In other words, one can plainly distinguish an explicit pair of agents (source and destination) according to communication purposes for semantic-oriented and goal-oriented communications. In contrast, semantic-aware communication ties a task that has to be performed jointly by several agents. In this way, multiple agents cooperate to complete a task in a centralized or distributed way through semantic-based communications. Semantic-aware communication differs from semantic-oriented and goal-oriented communications by approaching the semantic information as a result of agent behavior in the environment instead of extracting it from a data source.

We can give the following example. A base station communicates with several users spread over a given area. The base station tries to predict the conditions, i.e., the semantic information, of the mobile radio channel between itself and all users. The base station sets the schedule of broadcasts based on its channel forecast. Furthermore, there are case studies with multiple intelligent agents in which the semantic information extracted is specifically used to facilitate multi-agent cooperation, regardless of the task or goal of the agents themselves [9].

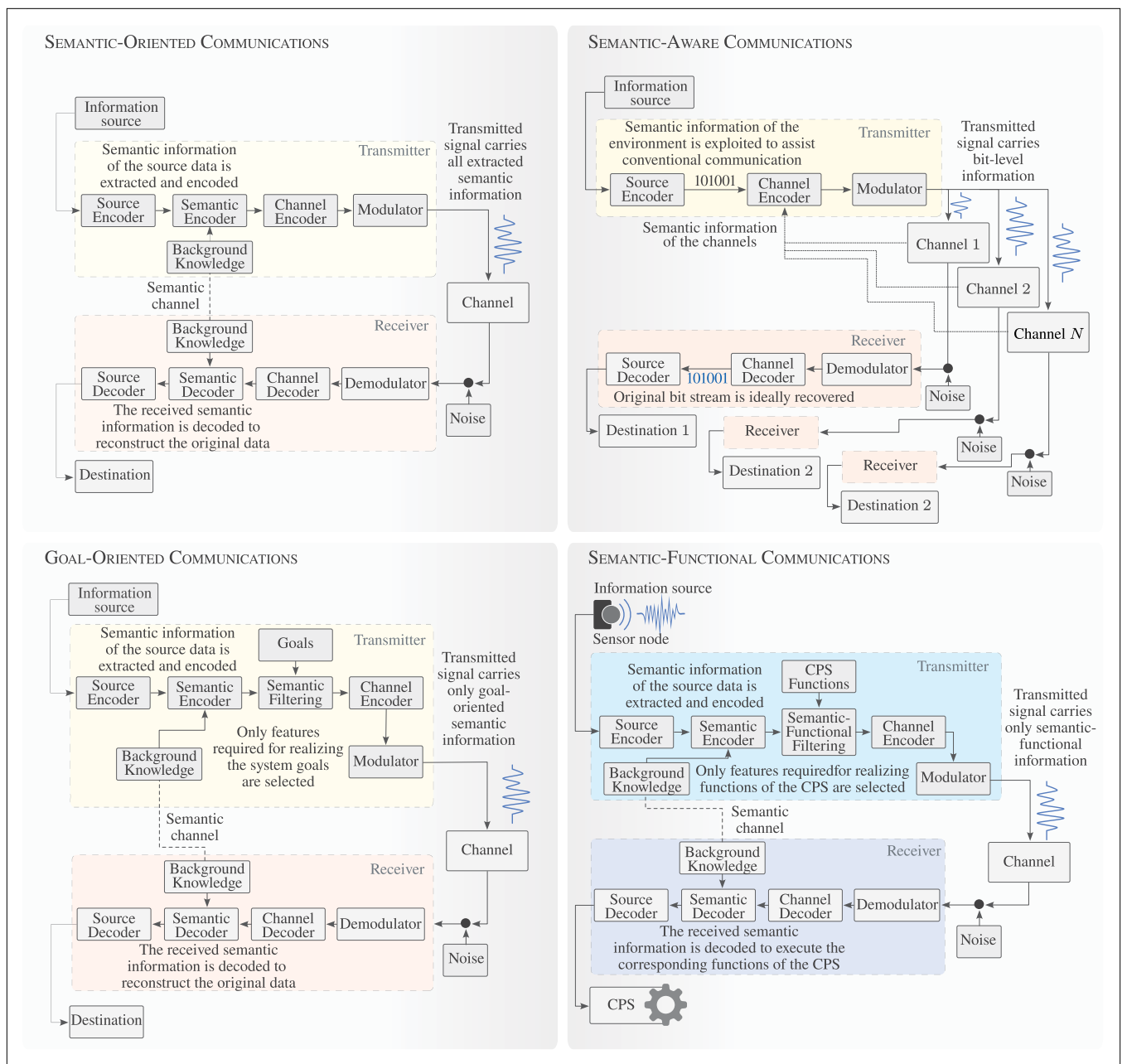


FIGURE 1. Comparison between different classes of semantic communications.

SEMANTIC-FUNCTIONAL COMMUNICATIONS FOR CPSs

As briefly introduced in Section II, the term *semantics* has a broad range of meanings in the wireless communication literature, dealing with aspects like information extraction, encoding, and contextualization. Our use of the term refers to the understanding of raw data as information within a given context, system, or end application. To avoid confusion, we define semantics as the outcome of a symbolic operation that assigns a purified (no misunderstanding) meaning in the cyber domain to data that have the potential to be directly or indirectly acquired from the physical world. For a given CPS, it is possible to further specify the semantics by determining the subset of all acquired data that support the CPS to

function within its operational limits; this subclass is named here as *semantic-functional*. Therefore, by semantic-functional, we mean semantics that enable the CPS to perform its function.

Although our definition of semantic-functional and pragmatic information—at least as the recent literature proposes it as in [12] that draws a parallel between effectiveness and pragmatics—are somewhat similar, we decide to keep the term semantic-functional because of the origin of the term pragmatics. The fields of linguistics and philosophy pioneered the definition of pragmatics, which can be briefly summarized as follows. Pragmatics has to do with the relations of signs to those who use the signs [13]. For instance, inferring the veracity of a statement under ordinary circumstances concerns pragmatical rules; when one hears a sentence, he/she inherently assumes that the speaker believes that sentence,

even though the speaker has not explicitly made it clear that it turns out to be true. Thus, a lie can be defined only by the pragmatic dimension of semiosis. Therefore, pragmatics does not directly deal with the effect of the message on system functionality; in our understanding, the use of pragmatic information in the recent communication theory would be better named practical information.

Let us illustrate the previously introduced concepts of semantic and semantic-functional. Assume that the temperature must be kept below a threshold by controlling the window in a room. The physical world, in this example, refers to the gases inside the room, window, walls, etc. At the same time, the temperature is an abstract concept that measures a statistical average of the level of agitation between molecules. By defining a scale (Kelvin, Celsius, etc.) for the mean agitation of molecules, we are, within our context, attributing semantics to a physical phenomenon and the data acquired therefrom. Furthermore, this purified concept of mean agitation is semantic-functional in our case because it enables the CPS to operate. Note that one can obtain different degrees of freedom from our semantic-functional measurements. For example, one can measure the temperature with a few tenths of precision or acquire a binary variable that expresses temperature being above or below the threshold; the most suitable design option depends on how strict are the functional requirements of the CPS in terms of desired room temperature. On the other hand, although a scale to measure the standard deviation of molecular agitation is semantic, it would not be semantic-functional because it does not have any role in the CPS operation.

Without loss of generality, let $a_i[k] \in \mathcal{A}$ be the i th attribute of a physical system Φ at a given (discrete) time $k \in \mathbb{N}^+$ and defined by a set of possible meaningful attributes $\mathcal{A}$. The series of the attribute $a_i[k-t], \dots, a_i[k] \forall i = \{1, \dots, N\}$ is the basis to determine the semantic-functional state $s_k \in \mathcal{S}$, where t is the time dependence, and $\mathcal{S}$ contains the possible semantic-functional states that Φ can experience, e.g. whether an attribute is outside of its operating range; this is defined as a history-dependent mapping (a semantic operation) from $f: \mathcal{A}^{N \times (t-1)} \rightarrow \mathcal{S}$. Note that both a_i and s_k are symbolically defined measurements, with the former being the data acquired from Φ and the latter being the semantic-functional representation of data in the cyber domain.

More specifically, let $X \in \mathcal{X}$ and $Y \in \mathcal{Y}$ represent the input and output messages of a communication channel, with $\mathcal{X}$ and $\mathcal{Y}$ corresponding to the input and output message sets, respectively. Also, we let the estimation of s_k in the decision-maker be denoted by $\hat{s}_k = r(Y)$, with $r: \mathbb{C} \rightarrow \mathcal{S}$ being a function to recover the semantic-functional state of Φ . Last, we have $X = m(s_k, s_{k-1}, \dots, s_{k-d})$ with

$m: \mathcal{S}^{d+1} \rightarrow \mathcal{X}$ being the transmission function that incorporates aspects of signal modulation, encoding, and medium access control, where d is the time dependence. It is noteworthy that non-transmission (silence or idle channel) is a valid state for X .

Let us consider the case where the physical system is not ideal and, indeed, diverges from the expected dynamics. In this case, we would have $\Pr[\hat{s}_k \neq s_k] > 0$ without explicit communication. One can reduce $\Pr[\hat{s}_k \neq s_k]$ as much as desired by performing explicit communication. However, we aim to minimize explicit communication—even though we have $\Pr[\hat{s}_k \neq s_k] > 0$ —while keeping the physical system operating.

In this context, the challenges of SFC are: (i) determining the semantic-functional states $s \in \mathcal{S}$ for the operation of the cyber-physical system; (ii) proposing an event-driven communication system to minimize the need for explicit communication; and (iii) defining what the acceptable level of errors that the SFC requires is, so that the cyber-physical system is still functional considering the proposed data transmission scheme.

The mathematical formulation of one of the simplest possible SFC problems can be stated as follows: the aim is to minimize the average occurrence of transmissions $\mathbb{E}[\Theta[k]]$ so that the performance indicator $\mathcal{V} < \delta$ is consistently met. $\Theta: \mathcal{X} \rightarrow \{0, 1\}$ is an indicator function that takes on the value 0 if the current semantic-functional state s must not be transmitted, and 1 otherwise, $\mathcal{V}$ refers to a given performance indicator of the CPS in consideration, and δ is the maximum acceptable value of $\mathcal{V}$ so that one can operate a given function of the CPS.

ILLUSTRATIVE EXAMPLE

In this section, we illustrate the potential of the proposed solution, considering a remote sensing task to be performed by a swarm of drones that need to be coordinated via wireless communications. It is worth indicating that this scenario, which is motivated by [14], is used here as an example to demonstrate the benefits of the proposed SFC approach to control the trade-offs that ultimately determine the performance of this CPS operation.

SCENARIO DESCRIPTION

The target is to monitor a specific area by performing remote sensing by a swarm of nine autonomous drones, which are assumed to be able to monitor such an area satisfactorily. To this end, the area is divided into nine square cells, assigning one drone to each cell. Figure 2 outlines a simplified illustration of one cell for each of the three proposed approaches: $\mathcal{S}_1$, $\mathcal{S}_2$, and $\mathcal{S}_3$. The drones are programmed to perform a random walk and return to the central position of their coverage area when they receive a return command indicating that they are outside of their specific region. After returning to the center, the drone restarts its random walk. For simplicity, we assume that at a given flight altitude, the area monitored by a drone at a given instant corresponds to $1/h$ of a cell, with $h > 1$ being a constant. Additionally, a drone can scan only the area directly beneath it, and all drones fly at the same altitude. Thereby, we can abstract from our

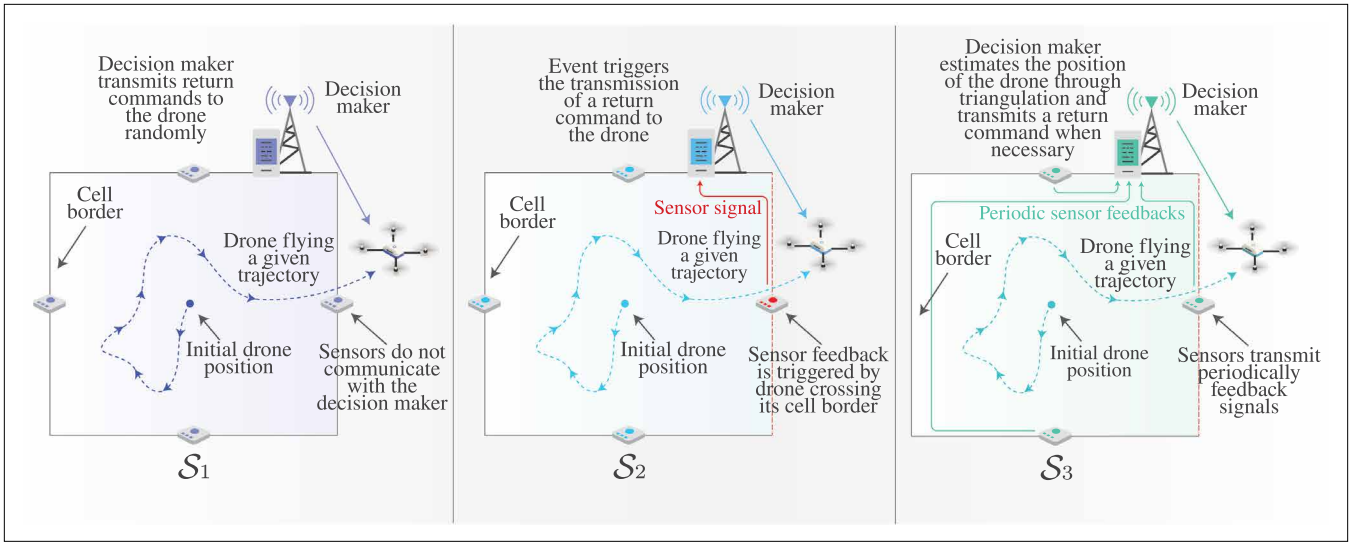


FIGURE 2. Simplified illustration of the three proposed solutions: $\mathcal{S}_1$, $\mathcal{S}_2$, and $\mathcal{S}_3$.

analysis the implications of the drone flight altitude, which would add unnecessary complexity to our explanation.

The purpose of our communication system is to enable the nine drones to stay within their respective areas minimizing explicit communications while controlling the performance indicator $\mathcal{V}_{\text{time}}$ and $\mathcal{V}_{\text{violation}}$ so that their value does not exceed its maximum acceptable upper bound. In our scenario, we have $\mathcal{V}_{\text{time}} = \{t_1, \dots, t_9\}$ and $\mathcal{V}_{\text{violation}} = \{v_1, \dots, v_9\}$, where t_i and v_i are the average time required for the i th drone to cover its respective area and the percentage of time the i th drone remained *outside* its respective area (perimeter violation), respectively. Note that the system allows, within certain limits, a drone to leave its area or overly concentrate navigation in the center of its coverage area. Both cases refer to inadequate coverage and lead to an increase in the average time needed to cover the entire area. The remote sensing task (e.g., what is monitored, how and to whom the drones will report their measurements) is beyond the scope of this work.

We assume the decision-maker can transmit the return command to drones through an error-free broadcast communication. A series of protocols and modulation standards can be listed as candidates for such communication; however, it is enough for our explanation to assume that the drones can receive the return command with negligible delay and that the decision-maker has unlimited access to the control channel of the drones. In addition, we have separate channels for control and sensor communication.

SEMANTIC-FUNCTIONAL COMMUNICATION

Our first challenge is to determine the semantic-functional states for the operation of the cyber-physical system under $\mathcal{V}_{\text{time}} < \delta_{\text{time}}$ and $\mathcal{V}_{\text{violation}} < \delta_{\text{violation}}$. Note that δ_{time} and $\delta_{\text{violation}}$ can represent either a physical constraint or a quantitative assessment of system functionality. For example, δ_{time} can be the maximum battery time of a drone. We propose three solutions as follows.

1. $\mathcal{S}_1 = \emptyset$, no system state is considered. This is an extreme case where the system is not monitored and there is no communication from sensors to the decision-maker. However, the decision-maker continues to act on the drones by randomly sending a return command to each drone.
2. $\mathcal{S}_2$ is constituted by the state that the system assumes regarding whether the drones are within their respective coverage areas. For example, a state would be all drones inside their respective subareas, another semantic-functional state would be to have just the first drone outside its respective subarea, and so on.
3. $\mathcal{S}_3$ has all possible combinations of the position (coordinates) of each drone. In other words, $\mathcal{S}_3$ refers to the drone positions for a given accuracy scale. For instance, a state would be all drones in their respective starting positions (center of subarea).

Let us propose a communication system in order to minimize the need for explicit communication from sensors to the decision-maker. Among some practical solutions to support system operation, we can highlight two: *event-triggered communication* for $\mathcal{S}_2$ and *periodic communication* for $\mathcal{S}_3$.

Regarding $\mathcal{S}_2$, an excellent approach would be to transmit when the drone crosses the border. For each sensor node, two events are possible: a drone crosses the border from right to left (or from top to bottom) or vice versa. As shown in [5] for event-triggered transmissions, the SFC technique overcomes orthogonal and random protocols for media access control regarding $\Pr[\hat{s}_k \neq s_k]$. Furthermore, this transmission scheme incorporates the process of modulation and encoding. Note that the SFC technique of [5] addresses how the sensors access the medium, but not the temporal dependence required by transmission function m .

The technique proposed in [5] ensures that the decision-maker is aware of the event even when collisions (two or more sensors attempting to transmit simultaneously) occur during a

transmission. On the other hand, such a method entails the possibility of false positives; however, one can have a false positive probability as low as desired. As we will see in more detail when discussing the numerical results, such a disadvantage from the communication point of view does not interfere with the proper functionality of the system. In contrast, false negatives—resulting from a collision if ALOHA were adopted, for example—would substantially degrade the system’s functionality.

For $\mathcal{S}_3$, a possible approach in terms of periodic communication would be as follows. We could deploy three sensor nodes so that the specific position of any drone can be obtained by triangulation. Thus, each of the three sensor nodes would periodically send the distance between itself and the drones. In this way, the decision-maker would be able to estimate the position of each drone and, if necessary, send a return command.

RESULTS

Let us assume a total monitoring area of 60 m × 60 m, resulting in nine cells of 20 m × 20 m. In this scenario, we consider the trade-offs involving the use of implicit/explicit communications, measured by $\Theta[k]$, and the performance $\mathcal{V}_{\text{violation}}$ and $\mathcal{V}_{\text{time}}$ that evaluate coverage. The reader is encouraged to scrutinize the details of the simulations in the Matlab files made available on GitHub.¹ The main parameters of our illustrative example are presented in Table 1.

We will now turn our attention to the minimization of the use of explicit communication $\Theta[k]$. Given the three cases discussed in Section IV-B, we have $\Theta_1[k] = 0$, $0 < \mathbb{E}[\Theta_2[k]] < 1$, and $\Theta_3[k] = 1$ for $\mathcal{S}_1$, $\mathcal{S}_2$, and $\mathcal{S}_3$, respectively. Therefore, we can say that the composition of semantic-functional states ($\mathcal{S}_1$, $\mathcal{S}_2$, and $\mathcal{S}_3$) defines $\mathbb{E}[\Theta[k]]$. Regarding $\mathcal{S}_1$, it is noteworthy that (i) if the frequency of the return command is high, the drones will tend to stay too close to the center in addition to overloading the communication system; and (ii) if the time between two subsequent return commands for the same drone is too long, it will tend

to not remain inside its cell. It is assumed that the discrete interval k lasts T_k seconds.

The communication for the second scenario is addressed as follows. By setting the time dependence as $d = 1$ and transmitting according to the SFC technique of [15] only when $s_k \neq s_{k-1}$, we define transmission function m . In this way, the sensing channel remains idle whenever $s_k = s_{k-1}$. Furthermore, the decision-maker can always estimate correctly the semantic-functional state at instant k by setting $\hat{s}_k = \hat{s}_{k-1}$ whenever no sensor has transmitted. The same value of transmission energy expenditure $\mathcal{E}$ is adopted for the three cases. Furthermore, we assume that a transmission, starting from both the sensor nodes and the decision-maker, takes $T_k = 0.1$ seconds for the three cases. Regarding the communication between the sensors and the decision-maker, note that $\Pr[\hat{s}_k \neq s_k]$ is inversely proportional to $\mathcal{E}$ exclusively for $\mathcal{S}_2$, whereas $\mathcal{S}_3$ requires $\mathcal{E}$ to be greater than a minimum operating value. This minimum value of $\mathcal{E}$ for $\mathcal{S}_3$ can be obtained through the Shannon’s channel capacity. We obtain $\mathbb{E}[\Theta_2[k]] = 0.0577$ per second and $\Pr[\hat{s}_k \neq s_k] = 2 \times 10^{-3}$ for $\mathcal{S}_2$ with a given set of drone random walk parameters.

The third hypothetical scenario has only three sensor nodes that constantly report their readings to the decision-maker, as briefly discussed above. In this approach, only one sensor node can transmit at a time. The sensor nodes update their readings to the decision-maker using the TDMA technique. A TDMA slot has T_k seconds—maintaining consistency with proposed solutions—and can convey information (distance between drone and sensor node) about only one drone. Hence, the drone position is updated at the decision-maker side each $27T_k$. We further assumed a random delay in the transmission of the return command; this delay reflects the time required to process the triangulation. Note that the decision-maker notices a drone outside its respective area with a delay of at least $3T_k$ and a maximum of $27T_k$, excluding the triangulation processing time. Accordingly, we assume that the time to decision-maker detects a drone outside its respective

Parameter description	Value	Applicable to
Total coverage area	60 m × 60 m	$\mathcal{S}_1$, $\mathcal{S}_2$, and $\mathcal{S}_3$
Size of a drone’s cell	20 m × 20 m	$\mathcal{S}_1$, $\mathcal{S}_2$, and $\mathcal{S}_3$
Probability of false alarm	$\Pr[\hat{s}_k \neq s_k] = 2 \times 10^{-3}$	$\mathcal{S}_2$
Timeout	uniformly distributed in the interval [2, 5] seconds	$\mathcal{S}_1$
Delay to detect a drone outside its cell	uniformly distributed in the interval [0.3, 2.7] seconds	$\mathcal{S}_3$
Maximum velocity of the drone	10 m/s	$\mathcal{S}_1$, $\mathcal{S}_2$, and $\mathcal{S}_3$
Length of the random walk	10^6 steps	$\mathcal{S}_1$, $\mathcal{S}_2$, and $\mathcal{S}_3$
Runs of the random walk	100	$\mathcal{S}_1$, $\mathcal{S}_2$, and $\mathcal{S}_3$
Transmission time	$T_k = 0.1$ seconds	$\mathcal{S}_1$, $\mathcal{S}_2$, and $\mathcal{S}_3$
Discrete interval for the simulation steps	$k = 0.1$ seconds	$\mathcal{S}_1$, $\mathcal{S}_2$, and $\mathcal{S}_3$
Area monitored by a drone at a given instant	$1/h = 0.0205$ of a cell	$\mathcal{S}_1$, $\mathcal{S}_2$, and $\mathcal{S}_3$

TABLE 1. Summary of main simulation parameters.

¹ <https://github.com/pedro-goria/Semantic-Functional-Communications>

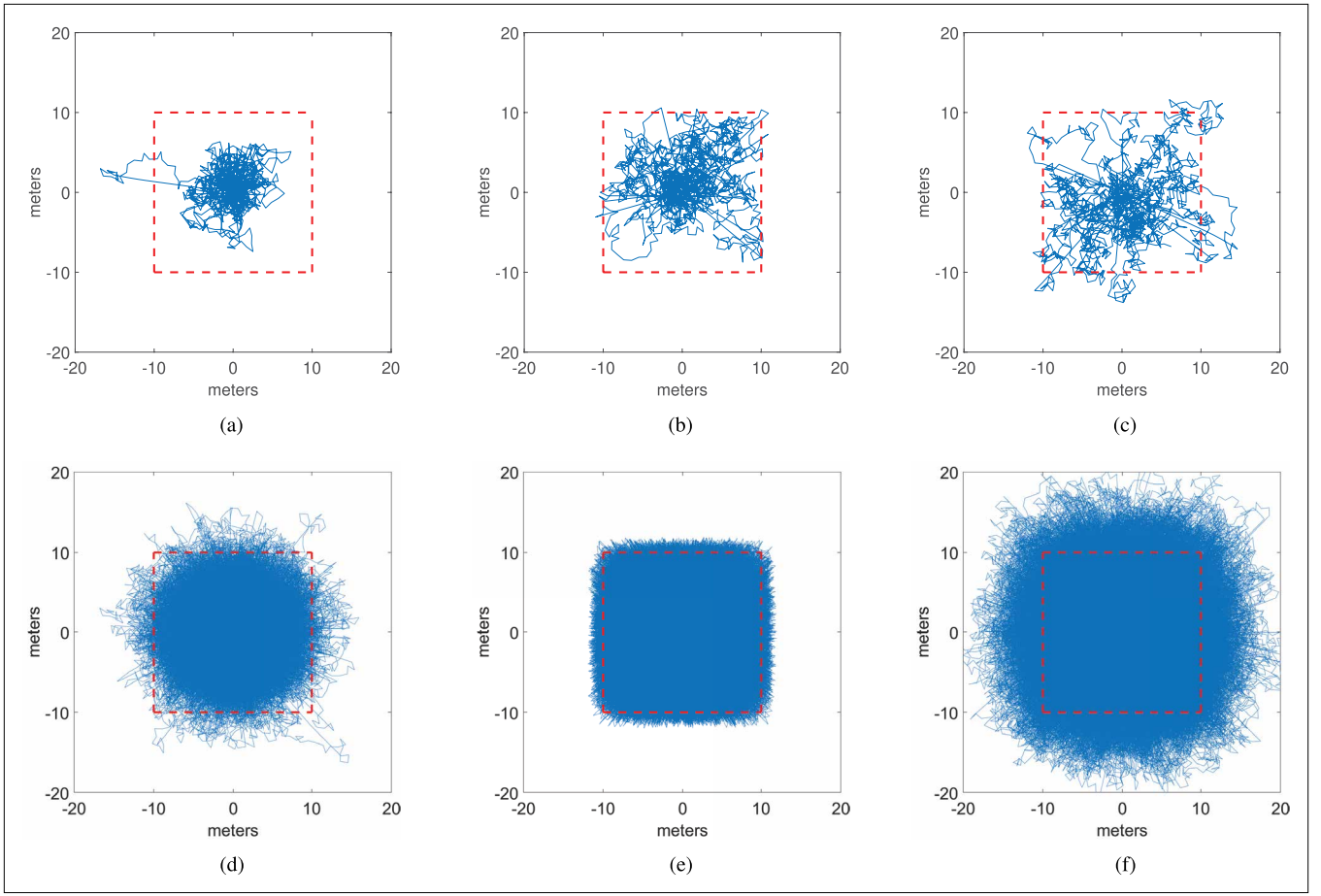


FIGURE 3. Comparison between the semantic-functional approaches a) and d) $\mathcal{S}_1$ with $\mathbb{E}[\mathcal{V}_{\text{time}}] = 71.44$ minutes and $\mathbb{E}[\mathcal{V}_{\text{violation}}] = 0.35\%$, b) and e) $\mathcal{S}_2$ with $\mathbb{E}[\mathcal{V}_{\text{time}}] = 9.04$ minutes and $\mathbb{E}[\mathcal{V}_{\text{violation}}] = 1.08\%$, and c) and f) $\mathcal{S}_3$ with $\mathbb{E}[\mathcal{V}_{\text{time}}] = 7.69$ minutes and $\mathbb{E}[\mathcal{V}_{\text{violation}}] = 7.55\%$. The solid blue line and the red dashed line represent the route taken by the drone and the boundary of its coverage area, respectively.

area follows a uniform distribution between $3T_k$ and $27T_k$ seconds.

Figure 3 outlines the results for these three hypothetical scenarios. The figure presents the systems (a) and (d) without using explicit communication, i.e., $\mathcal{S}_1$; (b) and (e) minimum explicit communication with the SFC representing $\mathcal{S}_2$; and (c) and (f) $\mathcal{S}_3$, which has maximum explicit communication. Furthermore, (a), (b), and (c) show the path of a drone after a short period of $2T_k \cdot 10^3$ seconds, while (d), (e), and (f) show the path of a drone after a long period of $2T_k \cdot 10^6$ seconds. Without explicit communication (i.e., the decision-maker operates blindly), we obtain $\mathbb{E}[\mathcal{V}_{\text{time}}] = 71.44$ minutes and $\mathbb{E}[\mathcal{V}_{\text{violation}}] = 0.35\%$ for an interval between two consecutive return commands uniformly distributed between 2 and 5 seconds. For $\mathcal{S}_2$, we have $\mathbb{E}[\mathcal{V}_{\text{time}}] = 9.04$ minutes and $\mathbb{E}[\mathcal{V}_{\text{violation}}] = 1.08\%$. Lastly, for $\mathcal{S}_3$, which constantly accesses the channel, we have $\mathbb{E}[\mathcal{V}_{\text{time}}] = 7.69$ minutes and $\mathbb{E}[\mathcal{V}_{\text{violation}}] = 7.55\%$. Note that $\mathcal{S}_1$ is the approach that keeps the drones closest to the center, $\mathcal{S}_2$ provides the most uniform coverage, and $\mathcal{S}_3$ slightly outperforms $\mathcal{S}_2$ in average coverage time $\mathbb{E}[\mathcal{V}_{\text{time}}]$.

Let $\mathcal{E}_T$ be the average energy spent on transmission for a subarea—we have included both the return command transmission and the sensor transmissions—per unit time normalized with

respect to $\mathcal{E}$. As mentioned, we assume that both a return command and a sensor transmission require $\mathcal{E}$ Joules. Thus, the calculation of $\mathcal{E}_T$ can be performed by adding the average number of sensor transmissions per second with the average number of return commands per second. For instance, we have that the average sensor transmission is zero, and the average return command per second is 0.2857 (i.e., the inverse of the average time between the return commands sent to the same drone) for $\mathcal{S}_1$.

Figure 4 compares the three degrees of freedom for the semantic-functional states adopted in this work. Three evaluation parameters are presented: transmission energy expenditure $\mathcal{E}_T$, average coverage time $\mathbb{E}[\mathcal{V}_{\text{time}}]$, and percentage of violation (outside) of area $\mathbb{E}[\mathcal{V}_{\text{violation}}]$. Note that we are dealing with three distinct cyber domains (i.e., $\mathcal{S}_1$, $\mathcal{S}_2$, and $\mathcal{S}_3$) while the physical domain remains unchanged. Figure 4 shows the impact of the cyber domain on the physical one.

CONCLUSION

This work provides a new approach by leveraging implicit communication based on semantic knowledge in the design of CPSs. Specifically, our main target was not to maximize the efficiency of the radio resources based on contextual knowledge

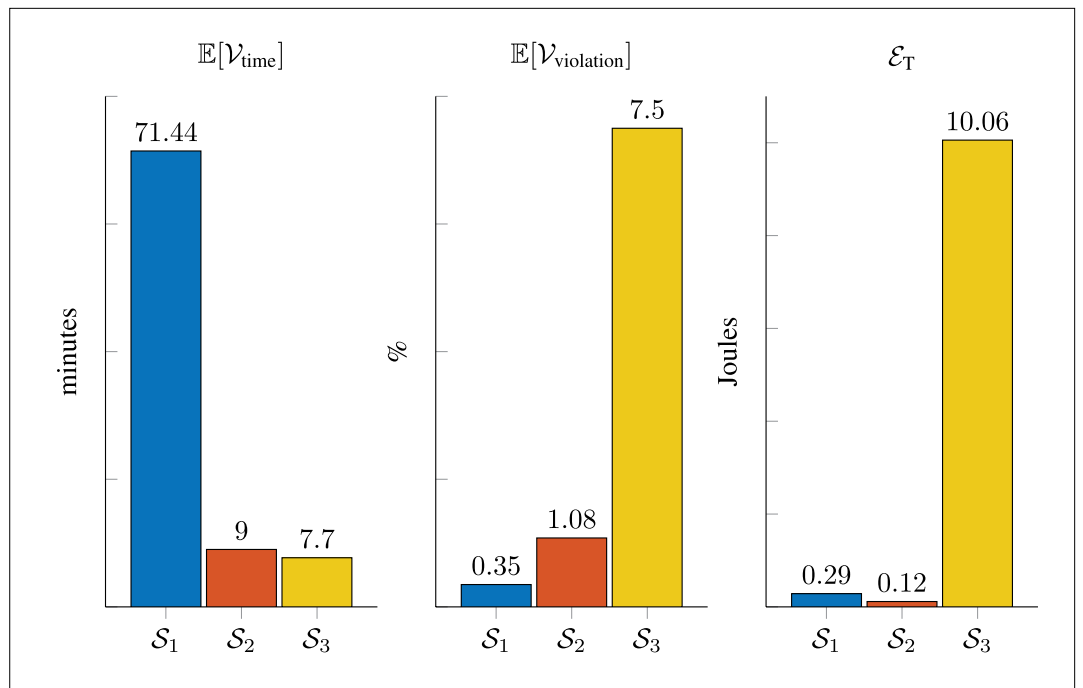


FIGURE 4. Comparison between the degrees of freedom to semantic-functional.

Our proposal paves the way for the standardization of sustainable networks, minimizing the use of resources and envisioning more tailored solutions for a given application.

but rather to minimize the use of such resources. At the same time, the CPS under consideration can still operate within the desirable performance range. With respect to the literature, we presented the formulation of what we call semantic-functional communication and successfully tested it in a CPS constituted by a swarm of nine drones that jointly perform remote sensing over a certain region. Our proposal paves the way for the standardization of sustainable networks, minimizing the use of resources and envisioning more tailored solutions for a given application. In addition, we investigate the impact of three distinct semantic-functional degrees of freedom on the system performance. Although our method is suitable for event-driven communications, a semantic-functional approach in systems that require periodic sampling is still an open challenge. The promising results we obtained set the path for a more organic collaboration between control and communication engineers, both to determine the fundamental limits of the operation of cyber-physical systems and to build more effective systems that use as few resources as possible to perform their particular tasks.

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BIOGRAPHIES

PEDRO E. GÓRIA SILVA (pedro.goria.silva@lut.fi) received the B.Sc. degree in telecommunication engineering in 2017, and the M.Sc. degree in telecommunications from the National Telecommunications Institute (INATEL), in 2020. He is currently pursuing the Ph.D. degree with the School of Energy Systems, LUT University, Finland, in partnership with the INATEL, Brazil. His research currently includes stochastic thermodynamics in communication, mobile communication, and fading channels.

PEDRO H. J. NARDELLI (Senior Member, IEEE) (pedro.nardelli@lut.fi) received the B.S. and M.Sc. degrees in electrical engineering from the State University of Campinas, Brazil, in 2006 and 2008, respectively, and the Ph.D. degree from the University of Oulu, Finland, and State University of Campinas following a dual degree agreement, in 2013. He is currently an Associate Professor (tenure track) in IoT in Energy Systems with LUT University, Finland. He holds a position of Academy of Finland Research Fellow.

ARTHUR S. DE SENA (Member, IEEE) (arthur.sena@oulu.fi) received the B.Sc. and M.Sc. degrees in teleinformatics engineering and computer engineering from the Federal University of Ceará, Brazil, in 2019 and 2017, respectively, and the D.Sc. degree (with distinction) in electrical engineering from LUT, Finland, in 2022. He is currently a Post-Doctoral Researcher in the 6G Flagship program with the University of Oulu, Oulu, Finland. Preceding this role, from 2022 to 2023, he was a Researcher at the AI and Digital Science Research Center, Technology Innovation Institute, Abu Dhabi, United Arab Emirates. From 2019 to 2022, he was a Junior Researcher with the Laboratory of Control Engineering and Digital Systems, Lappeenranta-Lahti University of Technology (LUT), Lappeenranta, Finland. He was awarded the Brazil Scientific Mobility Program scholarship, spanning from 2014 to 2015, in which he studied computer engineering as an Exchange Student at the Armour College of Engineering, Illinois Institute of Technology, Chicago, USA. His contributions to the field of wireless communication have garnered notable recognition, including the Nokia Foundation Award in October 2020, the LUT Research Foundation Award in December 2020, and the IEEE Global Communications Conference (GLOBECOM) Best Paper Award in December 2022. He has authored several peer-reviewed papers in prestigious journals and flagship conferences. He is a member of the IEEE

and the IEEE Communications Society. His research pursuits delve into the forefront of wireless communication and signal processing, encompassing ground-breaking subjects like intelligent reflecting surfaces, massive and holographic MIMO technologies, next-generation multiple access techniques, and the intersection with classical and machine learning-driven methods.

HARUN SILJAK (Senior Member, IEEE) (harun.siljak@tcd.ie) received the bachelor and master degrees in automatic control and electronics from the University of Sarajevo, in 2010 and 2012, respectively, and the Ph.D. degree in electrical and electronics engineering from the International Burch University, Sarajevo, in 2015. He is an Assistant Professor in embedded systems, optimisation, and control at the School of Engineering, Trinity College Dublin. Previously, he was a Marie Skłodowska Curie Fellow at TCD. His current research combine cyber-physical systems, complex networks, and unconventional computation and communications.

NIKO NEVARANTA (niko.nevaranta@lut.fi) received the B.Sc., M.Sc., and D.Sc. degrees in electrical engineering from the Lappeenranta University of Technology (LUT), Lappeenranta, Finland, in 2010, 2011, and 2016, respectively. He is currently an Assistant Professor of Industrial Informatics for Energy System Disruption and works at the Laboratory of Control Engineering and Digital Systems, LUT. His current research interests include data-driven modeling and control of energy conversion systems, system identification, parameter estimation, system monitoring, and diagnostics.

NICOLA MARCHETTI (Senior Member, IEEE) (nicola.marchetti@tcd.ie) received the Ph.D. degree in wireless communications from Aalborg University, Denmark, in 2007, the M.Sc. degree in electronic engineering from the University of Ferrara, Italy, in 2003, and the M.Sc. degree in mathematics from Aalborg University, in 2010. He is an Associate Professor in wireless communications at Trinity College Dublin, Ireland. He performs his research under the Irish Research Centre for Future Networks and Communications (CONNECT), where he leads the Wireless Engineering and Complexity Science (WhyCOM) Lab. He has authored in excess of 160 journals and conference papers, two books and nine book chapters, holds four patents, and received four best paper awards. His research interests include autonomous and self-organising networks, communications for biology, PHY and MAC for future networks, and quantum communications. He is a Fellow of Trinity College Dublin.

RAUSLEY A. A. DE SOUZA (Senior Member, IEEE) (rausley@inatel.br) received the B.S.E.E. and M.Sc. degrees from the National Institute of Telecommunication (INATEL), Brazil, in 1994 and 2002, respectively, and the Ph.D. degree from the State University of Campinas, Campinas, Brazil, all in electrical engineering. Prior to joining the Academy, he was with industry. In 2002, he joined INATEL, where he is currently a Full Professor. His research focuses on wireless communications.